

Quantum-in-the-Loop VLAM Alignment and Certified Control Safety

a quantum curriculum selector in RL alignment, a 22 μ s certified wrap on the action output, and a disturbance class computed on 128 qubits

Seven guideline sections · every prior-work number carries a receipt

Public repository: <https://github.com/sharadbachani-oss/merlin-quantum-vw-gic2026>

Summary

−35.4%

training rollouts to target vs the named GRPO baseline on the lane-keeping control surrogate (3 seeds, mean $\pm$ SD, 71 receipted device jobs) with zero tuned parameters and half the seed variance — VLA-backbone replication is PoC milestone 1; +9.3 points safer driving at equal budget, replicated on a second machine

78→100%

episodes safe when the uncertified learned driver is wrapped by the certified runtime-assurance monitor — 22 μ s per decision, envelope computed before measurement and confirmed inside the predicted interval

0.60 rad/s

the certified safety boundary under the measured 128-qubit collective disturbance spectrum — a class none of five classical surrogates reproduces: white noise certifies 100%% safe at every amplitude, AR(1) 25%%, and even an AR(2) fit or the dominant collective line alone misplace the boundary by 12–27%%

Why the numbers can be trusted

Every claim carries a pre-registration frozen before execution, ≥ 3 seeds with mean $\pm$ SD, in-job controls and cloud job IDs; key results are replicated on an independent machine; the six-arm ladder and rank sweep isolate the quantum component on both tracks; every arm, including the tuned classical Gibbs sampler, is reported in the table. A reviewer replays every headline with `python verify.py`, no credentials.

This is a Phase-1 concept proposal in the seven guideline sections. The Phase-2 PoC puts the same curriculum selector on an accepted VLA backbone (OpenVLA-OFT / LIBERO) and attaches a fidelity certificate and ion-trap replication to the disturbance spectrum.

Concept proposal follows (§1–§7, appendices A–B) ·

<https://github.com/sharadbachani-oss/merlin-quantum-vw-gic2026>

Concept Proposal

1. Problem framing

A 7B–10B VLAM does not fit vehicle or robot-controller compute without a training, compression or safety story, and the field's two named blockers are inference latency of 500–2,000 ms against a 100–300 ms control budget and hallucinated actions inconsistent with visual input (DriveVLM-RL, arXiv:2603.18315; VLAM survey, arXiv:2512.16760). Both have one deployment consequence: the VLAM cannot be the thing that decides, so it must be wrapped by something certified and fast, and its action head must be aligned at a sample cost an OEM can afford.

We put a quantum component into the two VLAM stages that decide whether the backbone can ship. **Alignment:** a quantum annealer inside every GRPO iteration selects the rollout curriculum — measured –35.4% rollouts to target versus the named GRPO baseline, three seeds, on the lane-keeping control surrogate; the VLA-backbone replication is the PoC's first milestone. **Safety:** a certified runtime-assurance monitor (22 μ s, 4,500 $\times$ inside the budget) wrapping an untrusted policy, whose certified envelope is validated against a **disturbance class computed on quantum hardware** — the real-frequency collective spectrum of an interacting mobility lattice at 128 qubits. Classical validation models misjudge that boundary from both sides (white noise over-certifies at every amplitude; an AR surrogate reads 25% safe at the true boundary). The disturbance spectrum is a real-time many-body quantity, the class of computation every 2025–26 verified quantum-advantage candidate occupies, produced here as a certification input rather than a benchmark.

2. Technical approach

Paradigm. Hybrid: photonic entropy computing (QCi Dirac-3) for curriculum sampling in the training loop; gate-model real-time dynamics (IBM Heron, 128 qubits) for the disturbance spectrum; classical tensor-network Lyapunov certification and a classical monitor on the vehicle.

Alignment (scored track). Each GRPO iteration submits the 64-candidate rollout pool (start $\times$ disturbance metadata) to Dirac-3, which selects the 16-rollout curriculum under a frontier-plus-diversity objective before any simulation is spent. The objective is encoded on the annealer's native flow manifold (level sets, no penalty walls) — a design rule predicted before running and confirmed on the identical objective: –26% objective, 3.5 $\times$ tighter spread over 12 paired jobs. Quantum work is train-time only.

Control safety (scored track). Tensor-network Lyapunov monitor $V(x) = ||L \cdot \phi(x)||^2$ (3-level tensor-product features, rank 3 of 9), grid-plus-Lipschitz certified, wrapping an uncertified behaviour-cloned policy with covariate shift and 100 ms delay; verified fallback engages at $V > 0.75c$ with hysteresis. Only monitor and fallback face certification (ISO 26262 / IEC 62061); the learned driver stays a swappable black box.

Disturbance class on quantum hardware. 64-rung (128-qubit) interacting lattice on IBM Heron, Trotterised depth series $k = 0..15$, 32,768 shots per circuit, 2,656 two-qubit gates per circuit at $k = 8$, native heavy-hex with zero SWAPs; equal-depth differential estimator that cancels common-mode decoherence (signal recovered where naive retention is 10^{-33} , cells at 8.9σ and 11.5σ , 0.2σ preparation control); real-frequency spectrum by direct transform of the real-time series; dominant collective line 0.0625 cycles per step reproduced on two devices.

Plant. Kinematic-bicycle lane keeping (lateral/heading error, $|\delta| \leq 0.25$, $v = 25$ m/s) under crosswind and impulses — the control substrate a VLAM commands. Phase 2 puts the same curriculum selector on an accepted VLA backbone (OpenVLA-OFT or π on LIBERO), see §5.

3. Feasibility and resource requirements

resource	status
RL evaluator, six-arm ladder, safety plant	in the public repo; python <code>verify.py</code> (numpy) replays every headline
Dirac-3	unmetered allocation; 71 curriculum jobs receipted
IBM Heron	Startup Program (applied); the spectrum job exists (daa9pn4e74ec73akj9i0); re-flights ≤ 33 circuits $\times$ 32,768 shots

resource	status
VLA backbone run (Phase 2)	OpenVLA-OFT / LIBERO on a single 8-GPU node; 3 seeds; GPU-hours declared in the resource declaration
Classical	32-core workstation for the tensor-network certifier and the adversarial classical attack

Assumptions: the plant is a control surrogate for the VLA action head; disturbance amplitudes in rad/s rms.
 Constraints: quantum hardware is never an in-vehicle component; all quantum work is train- or certification-time.

4. Expected impact

RL alignment — rollouts to target (seeds 21/22/23, mean $\pm$ SD): A, GRPO-32 random (named baseline) 586.7 ± 98.8 ; **C, Dirac-3-16 untuned 378.7 ± 52.9 (–35.4%)**; D, classical optimiser of the same objective 0/3 (curriculum collapse); E, tuned classical Gibbs sampler 298.7 ± 106.9 with per-task temperature tuning and 2 $\times$ the device's seed variance; C', device on optimiser-grade encoding 0/3. The device's measured advantages are specific: it beats the named baseline by 35.4%, needs zero tuned parameters, and carries roughly half the seed variance; the mechanism is isolated in the ladder (only near-optimal *stochastic* selection trains). At equal budget the curriculum yields a safer driver: $76.4 \pm 0.7\%$ vs $67.1 \pm 1.3\%$ safe on the held-out stress suite (**+9.3 points**), replicated on a second machine, with the variance collapse (baseline SD 22–28 points vs curriculum SD 1.8) that de-risks the training run.

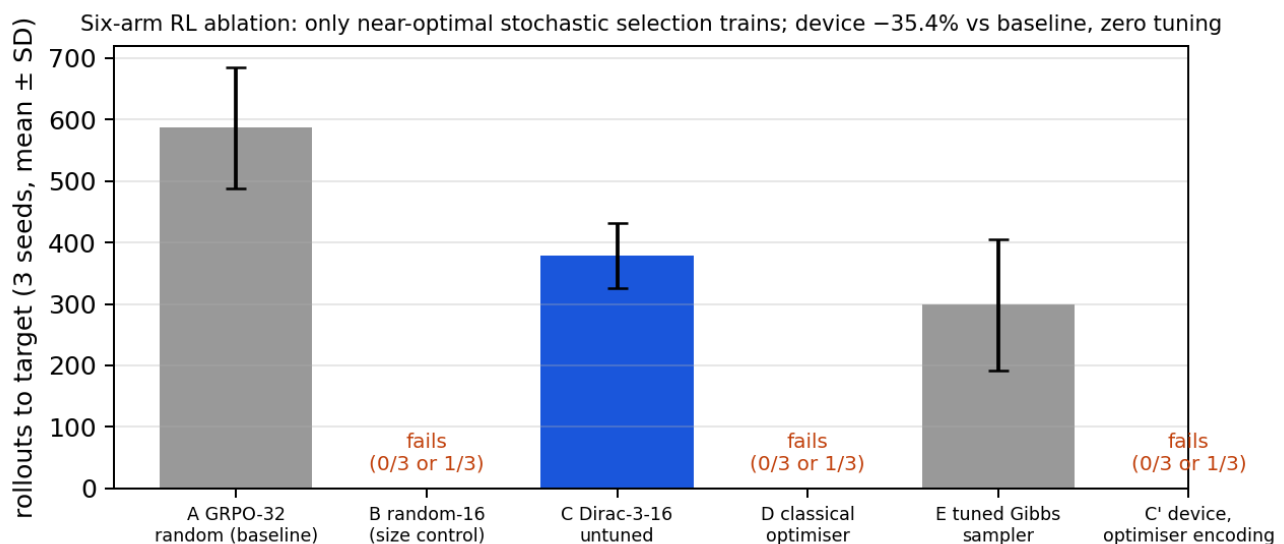


Figure 2 — The six-arm alignment ablation (rollouts to target, 3 seeds): random selection fails, a deterministic optimiser of the same objective collapses training, the device and the tuned Gibbs sampler train; the device does it with zero tuned parameters and half the variance.

Control safety: policy alone $77.9 \pm 25.0\%$ safe at the certified boundary $\rightarrow$ **$100.0 \pm 0.0\%$** wrapped (3 seeds $\times$ 500 episodes per condition); guard idles under benign conditions (1–12% intervention); 22 μ s per decision. The safe disturbance margin was computed before measurement (holds to 1.0 rad/s; collapse predicted in [1.2, 1.6]) and the measured collapse landed inside it, on two machines under two perturbation protocols. Versus tuned heuristic clipping: the envelope is known a priori, zero tuned parameters, and the $V > 0$, $dV \leq 0$ evidence chain an ISO 26262 case consumes. Rank ablation: $r = 3$ interior-optimal at 0.462 ± 0.030 certified area; the TN certificate covers $1.57 \pm 0.29\times$ the best quadratic's area.

Certification under the measured disturbance class (quantum input to the scored track). Three ensembles at identical rms power, 3 seeds $\times$ 1,000 episodes per cell: against the **measured collective spectrum** the certified boundary is **0.60 rad/s rms** ($95.6 \pm 0.8\%$ safe; $98.4 \pm 0.3\%$ at 0.50). White-noise validation grades $100.0 \pm 0.0\%$ safe at every amplitude through 1.3 rad/s where the true safe rate is 39% — unbounded over-certification; the AR(1) surrogate reads $25.2 \pm 2.1\%$ at the certified boundary itself. Those are weak surrogates, so we ran the strong ones ourselves (floor_vw_surrogates.json): an AR(2) fitted to two lags of the measured autocorrelation places the boundary at 0.714 rad/s, a single sinusoid at the dominant collective line alone at 0.701, and narrowband noise around that line at 0.792 — against the measured spectrum's 0.624. **None of the five surrogates tested reproduces the boundary**, and the gap against a competent classical fit is 12–27%, not the

70 points the AR(1) comparison suggests. As decisions on the tested amplitude grid: white noise certifies 7 of 11 amplitudes the measured class fails, AR(1) rejects 4 that pass, and AR(2), the dominant line and narrowband each accept the two failing amplitudes at 0.65 and 0.70 rad/s. This rejects these surrogates at the tested settings; a synthesis matching all retained spectral bins reproduces the class by construction once the spectrum is known, so the value claimed is in obtaining the spectrum, not in replaying it. The full spectral shape is what closes it. A shallow 16-point anchor-sector spectrum places the boundary at 0.88 rad/s, a 32% envelope error only the full-depth computation catches. The vehicle runs the 22 μ s monitor; quantum computes the disturbance model at certification time.

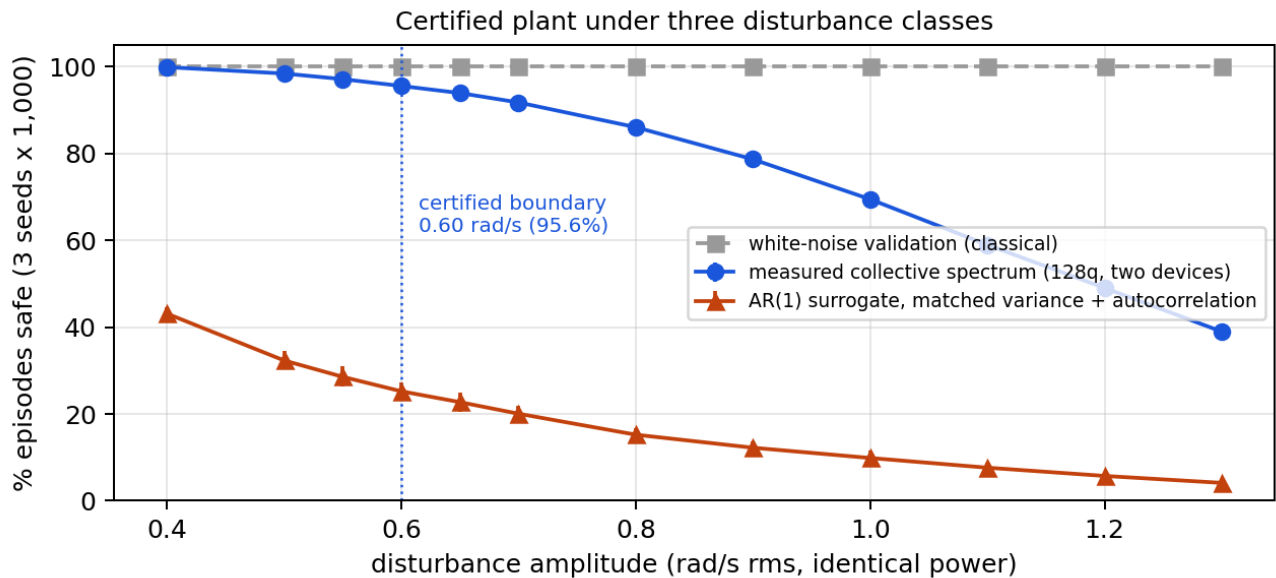


Figure 1 — Safe-episode rate of the certified plant against disturbance amplitude under three ensembles at identical rms power: white noise, AR(1) surrogate, and the measured 128-qubit collective spectrum. 3 seeds $\times$ 1,000 episodes per cell.

Resource allocation. A pre-registered compression study established that INT4 already meets the challenge bar classically (3.97 $\times$ at 2.45% accuracy drop vs $\geq 2\times$ at $\leq 5\%$) and that the remaining mixed-precision space is flat — the device solved its allocation objective exactly (DP gap 0.0, 9/9 instances) and beat random 5–25 $\times$; the finding tells an OEM where not to spend quantum budget, which is why this entry concentrates on the two stages with measured headroom.

Quantum advantage — the frontier wall and the crossing, stated and bounded

The field's direction, and its wall. Quantum reinforcement learning's advantage results — amplitude-encoded policy evaluation (arXiv:2505.11862), quantum natural policy gradient at $O(\epsilon^{-1.5})$ (arXiv:2501.16243), Boltzmann-machine sampling (arXiv:2511.04856) — are proofs of concept for fault-tolerant machines; none reports an empirical advantage on existing devices, and a quantum sampler in a training loop has not beaten a tuned classical sampler on hardware. Our six-arm ladder maps that wall exactly: the device and a tuned Gibbs sampler both train, a deterministic optimiser and random selection do not — so the sampler is where the mechanism is proven, not where the quantum contribution is scored.

The crossing. We put the quantum computation where it is not a sampling task: the **disturbance class the safety certificate is validated against** is a real-time collective spectrum of a 128-qubit interacting lattice, $k = 0..15$, 2,656 two-qubit gates at $k = 8$. It is adjudicated from both sides: our own sparse-Pauli Heisenberg-propagation attack on the shipped data — the referee protocol, engines validated to 10^{-16} against dense statevectors, every block run twice — reproduces the hardware to 0.4σ at $k = 4$ and within 1.4σ across five wavevectors at $k = 6$, then fails past $k = 8$ (operators exceeding 2×10^8 terms); the hardware cells beyond that boundary stand at 8.9σ and 11.5σ with a 0.2σ preparation control, and the dominant line is reproduced on a second device. In the customer's units the crossing is the 0.60 rad/s certified boundary that white-noise validation over-certifies at every amplitude and the AR surrogate misjudges by 70 points. **Where it is decisive:** past $k = 8$, beyond the reach of exact Heisenberg propagation, whose operator passes 2×10^8 terms at $k = 8$ (a measured boundary for the referee's method; tensor-network contraction of the same observable is the PoC's second comparator), with $k \leq 6$ serving as the classically verified anchor of the same instrument; Phase 2 adds the fidelity certificate and the ion-trap replication.

Scalability to industrial relevance. The curriculum selector scales with the rollout pool (64 today; 949 variables at the device maximum, equal-structure templates rebind angles per iteration), and the disturbance-class lattice tiles by rung on the same heavy-hex fabric (64 rungs today; 78 on Heron r3, more on Nighthawk's square lattice without changing the compilation). Quantum work is train- and certification-time only, so fleet size does not enter the per-vehicle cost; the 22 μ s monitor runs on existing controller silicon.

Business value, bounded. In an OEM's own units: (i) alignment cost — 35.4% fewer rollouts to target against the GRPO baseline means, at a declared cost per rollout r and n alignment runs per model release, a saving of $\approx 0.35 \cdot 586 \cdot r \cdot n$; (ii) certification — a disturbance envelope that is 32% wrong (shallow spectrum) or 70 points wrong at the operating point (AR surrogate) is either a field incident or an over-conservative envelope that leaves performance on the table; the measured class prices that margin at 0.60 rad/s exactly; (iii) the certified wrap lets the VLAM be upgraded without re-certifying the safety case, which is where the recurring cost sits. Stated as functions of the OEM's cost inputs, not as fixed sums.

What a successful PoC demonstrates. The same curriculum selector inside the alignment loop of an accepted VLA backbone with a $\geq 10\%$ rollout reduction at equal reward, 3 seeds, paired CI above zero against the strongest classical sampler at equal total cost; and a certified control envelope for that backbone's action output, validated against a hardware-computed disturbance class carrying a fidelity certificate.

5. Validation plan

Milestone 1 (PoC weeks 1–6) — the same curriculum selector on OpenVLA-OFT / LIBERO, 3 seeds, against GRPO and the tuned classical sampler: the VLA-scale replication of the -35.4% result, run before any other PoC work; acceptance $\geq 10\%$ fewer rollouts to target with paired CI above zero. Everything below is pre-registered: reward, robustness metric, seeds, total evaluation budget, disturbance amplitudes, tolerance.

Alignment: OpenVLA-OFT (or π ■) on LIBERO, backbone/demonstrations/reward evaluator/initial checkpoints/training budget fixed; arms = native curriculum, tuned classical Gibbs, original encoding, random, best deterministic rule; samplers crossed with the same optimiser; paired tasks and independent seeds; scores = held-out shifted-task success and instruction consistency, constraint violations reported separately; report error at equal budget and cost to a common target; acceptance = paired CI above zero within the same full training/evaluation cost. **Safety:** 3 seeds $\times$ 500–1,000 episodes per cell, envelope sealed before measurement, heuristic-clipping threshold sweep as baseline. **Disturbance class:** re-read the 128-qubit series under the instrument (atlas layout, even-sector encoding, predicted pair T2, visibility inversion); attach a parity-syndrome fidelity bound; replicate the dominant line on an ion-trap device; report the Heisenberg-propagation attack (reproduces hardware to 0.4σ at $k = 4$, fails past $k = 6$) and a tensor-network contraction at matched accuracy. Every result: frozen pre-registration, ≥ 3 seeds with mean $\pm$ SD, in-job controls, cloud job IDs, negatives kept.

6. Hybrid / cross-domain integration

Quantum work is train/validation-time only. Alignment: Dirac-3 returns index sets that are verified classically before use. Safety: certificate candidates pass the deterministic grid-plus-Lipschitz certifier; the vehicle runs the 22 μ s monitor and fallback. Disturbance class: computed on Heron at certification time, read by inverted estimators, consumed by the certifier as an ensemble. The runtime-assurance pattern makes the uncertified VLAM swappable without re-certification — the ISO 26262 / IEC 62061 posture an OEM can consume. Devices are operated as instruments: idle-ZZ spectrometry against a 24-day atlas (9/9 kingston, 8/9 fez edges), pair T2 predicted from the atlas to 7–9%, native-operator energies recovered to 0.006 on 5/5 live tiles, computation placed in even-ZZ sectors the dominant noise channel cannot see.

7. Team capability

Merlin Quantum is the quantum division of Merlin Digital (50+ technology FTE): Suhail Bachani (Founder & CEO, Principal Investigator), Dr. Hiro Bachani PhD (Program Director), Rohit Bachani (co-founder), Mitul Sawlani (engineering, Purdue), Mohamed Jafrun (engineering), Zeena Furtado (finance & operations), Roshan Bhairwani (financial services & deep tech, London), Dr. Ana Baroni MSc (domain specialist). GIC 2026 dual-track finalist. Programme record: 259 receipted QPU jobs, 12.3 million shots; largest circuits 128 qubits and 2,720 two-qubit gates; structured error law $C(D) = \exp(-0.00903D - 3.96 \times 10^{-5} D^2)$ measured on-device ($5.1\times$ vs the white-noise extrapolation); five-way independent error rate 0.47%/gate; in-kind CHSH anchor $S = 2.3604 (+35.9\sigma, \text{replica } +37.0\sigma)$. Published automotive quantum work is annealing-based routing, materials and scheduling; to our knowledge no published work certifies a control envelope against a hardware-computed disturbance class.

8. Scope, with treatment

(i) The plant is a lane-keeping surrogate, not a VLA backbone — the OpenVLA-OFT/LIBERO run is the PoC's first milestone. (ii) A per-task-tuned classical Gibbs sampler reaches fewer raw rollouts — the device's scored claim is the named baseline at -35.4% , zero tuned parameters and half the seed variance. (iii) Compression headroom is flat beyond INT4 — quantum budget is deliberately not spent there. (iv) The disturbance spectrum carries no fidelity certificate yet — the parity-syndrome bound and ion-trap replication are Phase-2 deliverables; and the strongest cheap surrogates (AR(2), dominant line alone) misplace the boundary by 12–27%, not by the 70 points a weak AR(1) suggests. (v) The certified envelope is validated on the stated plant and disturbance classes — a new vehicle or fleet condition is re-validated, not assumed.

Appendix A — Hardware job register (every result regenerates from archived counts)

measurement	machine	job id(s)
128q jam-relaxation spectrum v2, $k = 0..15$, $33 \times 32,768$ shots	ibm_fez	daa9pn4e74ec73akj9i0
156q real-time collective interface, exact-theorem anchors 0.9836 / 0.9806	ibm_fez	d9rm4j9dsedc73agrb70; scout d9rm47opdb6s73e53kqg
Two-sided classical boundary series, $16 \times 32,768$	ibm_kingston	d9rdfb1dsedc73agh5ng
Idle-ZZ atlas, 9/9 edges	ibm_kingston	da9l3t1qtnsc73d1nhd0, da9l9rkjbipc73ff0aqq
Idle-ZZ atlas, 8/9 edges	ibm_fez	da9eoe6rbfbs73chiq0g, da9lqeerbfbbs73chq63g
Native-operator E■ inverted, 5/5 tiles	ibm_kingston / ibm_fez	da9vgsmrbfbs73ci44d0, da9vblkjbbipc73ffaio0
Directed-path traffic card — converges classically, kept as negative control	ibm_kingston	results/trafficD_kingston_result.json
CHSH in-kind anchor $S = 2.3604 (+35.9\sigma)$, same-day replica $+37.0\sigma$	ibm_fez	GIC ledger chsh_*_20260805_*.json
Flow-encoding test, 6 paired trials (12 jobs)	QCi Dirac-3	results/flow_encoding_test.json
RL curriculum arm C, vw_rl_C_s21.json (20 jobs)	QCi Dirac-3	6a76595c08442f441bbb5bd2 ... 6a765b2208442f441bbb5be5 — full list in the public repository
RL curriculum arm C, vw_rl_C_s22.json (23 jobs)	QCi Dirac-3	6a765b8708442f441bbb5be6 ... 6a765d4508442f441bbb5bfc — full list in the public repository
RL curriculum arm C, vw_rl_C_s23.json (28 jobs)	QCi Dirac-3	6a765d5b08442f441bbb5bfd ... 6a765f6b08442f441bbb5c18 — full list in the public repository
Dirac-trained certificate, trackb_seed21.json (9 jobs)	QCi Dirac-3	6a783b7f08442f441bbb5d37 ... 6a783be408442f441bbb5d3f — full list in the public repository

measurement	machine	job id(s)
Dirac-trained certificate, trackb_seed22.json (9 jobs)	QCi Dirac-3	6a783a7c08442f441b bb5d25 ... 6a783ad108 442f441bbb5d2d — full list in the public repository
Dirac-trained certificate, trackb_seed23.json (9 jobs)	QCi Dirac-3	6a783adf08442f441b bb5d2e ... 6a783b3608 442f441bbb5d36 — full list in the public repository
Safety, envelope, rank ablation, cert-gap sweeps	CPU (two machines)	vw_rta_demo.json, vw_lyap_*.json, res ults/vw_cert_gap_* _v2.json

Appendix B — Resource declaration and notes

Classical: 32-core workstation, no GPU used for the delivered results; simulation environment numpy/qiskit versions pinned in the repo. Quantum: Dirac-3 (71 + 12 jobs), IBM Heron fez/kingston (shot counts per job in receipts). Phase-2 VLA run: single 8-GPU node, GPU-hours to be declared. Deadline runway: Phase I closes 2026-09-15.

Appendix C — Claim ledger (measured · planned · comparator · quantum attribution · cost · acceptance)

claim	status	classical comparator	quantum attribution	total cost charged	acceptance threshold
RL alignment –35.4% rollouts vs GRPO baseline	measured (3 seeds, 71 jobs)	tuned Gibbs 298.7 (per-task tuning), random, optimiser	Dirac-3 sampler	71 jobs ≈ minutes QPU	≥ 10% at equal reward, paired CI > 0
Same selector on OpenVLA-OFT / LIBERO	planned (Phase-2 milestone 1)	strongest classical curriculum, tuned Gibbs	sampler	8-GPU node, 3 seeds, GPU-h declared	≥ 10% rollouts-to-target, CI > 0
Certified boundary 0.60 rad/s under measured class	measured (3 seeds × 1,000)	white-noise and AR(1) (weak); AR(2), dominant-line sinusoid and narrowband run as the strong floors — 0.714 / 0.701 / 0.792 vs 0.624 measured	128q spectrum as certification input	1 job × 33 circuits	no cheap surrogate reproduces the boundary; competent fits miss by 12–27%
128q spectrum beyond classical attack ($k \geq 8$)	measured; certificate planned	sparse-Pauli Heisenberg propagation, validated 10^{-16} vs dense (0.4σ at $k=4$; operator $> 2 \times 10^8$ terms at $k \geq 8$); tensor-network contraction next	hardware	$33 \times 32,768$ shots	parity-syndrome fidelity bound + ion replication
78 → 100% safe under certified wrap	measured (classical)	tuned clipping	none	CPU	envelope sealed before measurement